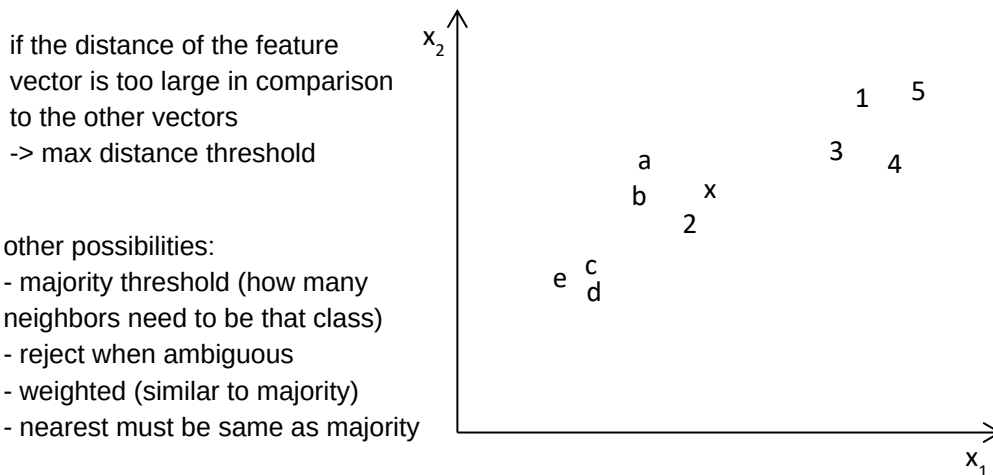




kNN-Classification – Exercise 2

The diagram below shows 2-dim feature vectors of the two classes “letter” and “digit” (displayed as letters a – e and digits 1 – 5, respectively). We will apply different forms of the nearest neighbor classifier for differentiating these classes, using Euclidean distance as metric. Which class is assigned to the unknown feature vector x in the following cases?

- Standard 1-NN classifier digit 100%
- 3-NN classifier letter 66%
- Give criteria for rejecting feature vectors using a 3-NN classifier. Which class is assigned to the unknown feature vector x now?



- A disadvantage of the k NN classifier is that the complete training sample must be stored in memory and searched during classification. Propose a method to avoid this. Are there any disadvantages to your method?

- create grid based on data points and evaluate neighboring grid cells instead of individual points (works for small k , similar quality)
- tree grid structure
- clustering of close data points, data reduction (reduces quality)
- pre-calculate decision boundary and evaluate data point based on boundaries (hard to apply)
- only save center of each class (extreme version of clustering) and check distance to those centers, very high bias, very low variance

moves away from laziness and pushes workload from inference to training